1. The fundamental architecture of modern digital computers based on stored-program concept was proposed by:

(A) John von Neumann
(B) Alan Turing
(C) Charles Babbage
(D) Blaise Pascal

2. The three primary hardware subsystems of a classical Von Neumann computer architecture are:

(A) CPU, Memory, and Input/Output (I/O) Interfaces
(B) Monitor, Keyboard, and Mouse
(C) Hard Disk, RAM, and Cache only
(D) Compiler, Assembler, and Linker

3. The Central Processing Unit (CPU) consists essentially of which two primary functional units?

(A) Arithmetic Logic Unit (ALU) and Control Unit (CU)
(B) ROM and RAM
(C) Modem and Router
(D) Flash Memory and Cache

4. The electronic clock speed of modern computer microprocessors is typically measured in:

(A) Gigahertz (GHz)
(B) Megabytes (MB)
(C) Nanoseconds (ns)
(D) Kilowatts (kW)

5. In computer memory hierarchy, which storage element possesses the fastest access time and lowest latency?

(A) CPU Registers
(B) L1 Cache Memory
(C) Main RAM
(D) Solid State Drive (SSD)

6. Which type of computer memory is volatile, losing its stored information immediately upon power interruption?

(A) Random Access Memory (RAM)
(B) Read Only Memory (ROM)
(C) Flash NAND Memory
(D) Magnetic Hard Disk Drive (HDD)

7. The system bus connecting the CPU to memory and peripherals is divided into:

(A) Data Bus, Address Bus, and Control Bus
(B) Serial Bus, Parallel Bus, and USB
(C) Optical Bus and Copper Bus
(D) Ethernet Bus and Token Ring

8. The width of the Address Bus determines which fundamental capability of a computing system?

(A) The maximum directly addressable physical memory capacity
(B) The execution clock speed
(C) The number of bits processed per cycle
(D) The data transfer rate

9. Data Communication is defined as the exchange of data between two devices via some form of:

(A) Transmission medium (wired or wireless channel)
(B) Operating system call
(C) Software compiler
(D) Instruction register

10. The five fundamental components of any data communication system are:

(A) Message, Sender, Receiver, Transmission Medium, and Protocol
(B) CPU, RAM, Hard Disk, Monitor, and Mouse
(C) HTTP, TCP, IP, Ethernet, and 802.11
(D) Client, Server, Router, Switch, and Bridge

11. In data communications, the agreed-upon set of syntax, semantic, and timing rules governing communication is a:

(A) Protocol
(B) Standard
(C) Operating System
(D) Firmware

12. A transmission mode in which data transfer occurs strictly in one fixed direction only (e.g., keyboard to CPU) is:

(A) Simplex mode
(B) Half-Duplex mode
(C) Full-Duplex mode
(D) Multiplex mode

13. A communication mode where both stations can transmit and receive data, but NOT at the same time (e.g., walkie-talkie) is:

(A) Half-Duplex mode
(B) Simplex mode
(C) Full-Duplex mode
(D) Dual-Simplex mode

14. A communication mode where both communicating devices can transmit and receive data simultaneously (e.g., telephone call) is:

(A) Full-Duplex mode
(B) Half-Duplex mode
(C) Simplex mode
(D) Asynchronous mode

15. The geometric or physical arrangement of nodes and connecting links in a computer network is termed its:

(A) Network Topology
(B) Protocol Stack
(C) Switching Matrix
(D) Routing Table

16. In which network topology is every node connected to every other node by a dedicated point-to-point physical link?

(A) Mesh Topology
(B) Star Topology
(C) Bus Topology
(D) Ring Topology

17. For a fully connected mesh network with n nodes, the total number of physical duplex communication links required is:

(A) n * (n - 1) / 2
(B) n * (n + 1) / 2
(C) n - 1
(D) 2 * n

18. In a Star Topology, each device is connected through a dedicated point-to-point link directly to a:

(A) Central Hub or Switch
(B) Terminator
(C) Token Ring
(D) Backbone cable

19. In a Bus Topology, all network devices share and tap into a single common communication channel called the:

(A) Backbone cable (with terminators at both ends)
(B) Central controller
(C) Mesh link
(D) Token coordinator

20. A major operational disadvantage of Bus Topology is that:

(A) A break or fault in the main backbone cable disrupts transmission across the entire network
(B) It requires excessive cable length
(C) It requires a complex central switch
(D) It cannot support broadcasting

21. In a Ring Topology, each device is connected to exactly two neighboring devices, and signals travel in:

(A) A single closed unidirectional (or bidirectional) loop with repeaters
(B) Random directions simultaneously
(C) Star formation to central hub
(D) Point-to-multipoint branches

22. A computer network privately owned and operated within a single office, room, or building spanning up to ~1 km is a:

(A) Local Area Network (LAN)
(B) Wide Area Network (WAN)
(C) Metropolitan Area Network (MAN)
(D) Global Area Network (GAN)

23. A network spanning an entire city, town, or campus (such as a municipal cable television network) is classified as a:

(A) Metropolitan Area Network (MAN)
(B) Local Area Network (LAN)
(C) Personal Area Network (PAN)
(D) Virtual Private Network (VPN)

24. A network spanning vast geographical distances across countries or continents (e.g., the global Internet) is a:

(A) Wide Area Network (WAN)
(B) Local Area Network (LAN)
(C) Storage Area Network (SAN)
(D) Personal Area Network (PAN)

25. A very short-range wireless network connecting personal devices within a range of ~10 meters (e.g., Bluetooth) is a:

(A) Personal Area Network (PAN)
(B) Wide Area Network (WAN)
(C) Local Area Network (LAN)
(D) Campus Area Network (CAN)

26. The world's first operational packet-switching computer network and the direct precursor to the Internet was:

(A) ARPANET (Advanced Research Projects Agency Network, 1969)
(B) NSFNET
(C) Ethernet
(D) ALOHANET

27. Which key international organization is responsible for publishing global telecommunications and networking standards (OSI model)?

(A) ISO (International Organization for Standardization)
(B) IEEE
(C) IETF
(D) W3C

28. The organization responsible for developing and publishing Internet standards (RFC documents) such as TCP and IP is:

(A) IETF (Internet Engineering Task Force)
(B) ISO
(C) ANSI
(D) ITU-T

29. Which metric measures the maximum theoretical data-carrying capacity of a communication channel in bits per second (bps)?

(A) Bandwidth
(B) Throughput
(C) Latency
(D) Jitter

30. The actual measured rate of successful data delivery over a communication link under real-world conditions is termed:

(A) Throughput
(B) Bandwidth
(C) Propagation delay
(D) Attenuation

31. The total time required for an entire message to travel from sender to receiver across a network is known as:

(A) Latency (or Total Delay)
(B) Jitter
(C) Throughput
(D) Bandwidth

32. Latency in computer networks consists of four distinct components: Processing Delay, Queuing Delay, Transmission Delay, and:

(A) Propagation Delay
(B) Modulation Delay
(C) Sampling Delay
(D) Switching Delay

33. Transmission Delay (T_tx) for a packet of length L bits transmitted over a link of bandwidth R bps is calculated as:

(A) T_tx = L / R
(B) T_tx = R / L
(C) T_tx = L * R
(D) T_tx = Distance / Velocity

34. Propagation Delay (T_prop) for a signal travelling distance d across a medium with signal propagation speed v is:

(A) T_prop = d / v
(B) T_prop = v / d
(C) T_prop = d * v
(D) T_prop = L / R

35. The variation in packet arrival delays in a packet-switched network, critical for voice and video streaming, is termed:

(A) Jitter
(B) Skew
(C) Attenuation
(D) Crosstalk

36. A network connection in which a dedicated, unshared physical path is established between nodes before communication begins is:

(A) Circuit Switching
(B) Packet Switching
(C) Message Switching
(D) Virtual Routing

37. In Packet Switching, discrete chunks of data containing header and payload travel independently across the network as:

(A) Packets (or Datagrams)
(B) Continuous streams
(C) Virtual circuits
(D) Analog carriers

38. Which switching paradigm is the foundational architecture of the modern Internet?

(A) Packet Switching (Store-and-Forward Datagrams)
(B) Circuit Switching
(C) Message Switching
(D) Space Division Switching

39. In computer networking, an intermediary hardware device that operates at the Data Link layer to forward frames based on MAC addresses is a:

(A) Network Switch (Bridge)
(B) Repeater
(C) Hub
(D) Passive splitter

40. An internetworking device operating at the Network layer that inspects destination IP addresses to route packets between disparate networks is a:

(A) Router
(B) Hub
(C) Repeater
(D) Modem

41. How many duplex communication links are required to interconnect 8 computers in a fully connected physical mesh topology?

(A) 28 links (8 * 7 / 2)
(B) 56 links
(C) 16 links
(D) 8 links

42. If 12 regional computing centers are to be interconnected using a Star topology with a central hub, how many cables and ports are needed?

(A) 12 cables and 12 hub ports
(B) 66 cables
(C) 24 cables
(D) 11 cables

43. A packet of size L = 1500 bytes is transmitted over a high-speed fiber link of bandwidth R = 1 Gbps (10^9 bps). The transmission delay is:

(A) 12 microseconds (12,000 bits / 10^9 bps)
(B) 1.5 microseconds
(C) 120 microseconds
(D) 1.5 milliseconds

44. A fiber optic undersea cable spans distance d = 6000 km between two continents. If light travels at v = 2 * 10^8 m/s in fiber, propagation delay is:

(A) 30 milliseconds (6,000,000 m / 2*10^8 m/s)
(B) 3 milliseconds
(C) 300 milliseconds
(D) 0.3 milliseconds

45. In a client-server architecture, which entity actively initiates contact by sending requests?

(A) The Client
(B) The Server
(C) The DNS Root
(D) The Gateway Router

46. In a Peer-to-Peer (P2P) network architecture, participating nodes (peers):

(A) Act simultaneously as both clients and servers with equivalent privileges
(B) Require a centralized master database server
(C) Communicate only via dedicated circuit lines
(D) Cannot share local files

47. The Bandwidth-Delay Product (BDP = Bandwidth * Round Trip Time) represents:

(A) The maximum volume of unacknowledged data that can fill the transmission pipe at any instant
(B) The packet loss probability
(C) The time required to establish a TCP connection
(D) The server processing overhead

48. For a transcontinental link with bandwidth R = 100 Mbps and RTT = 50 ms, the Bandwidth-Delay Product is:

(A) 5,000,000 bits (625 KB)
(B) 500,000 bits
(C) 2,000 bits
(D) 50 MB

49. Which standard body oversees the global coordination of the DNS Root, IP address allocation, and autonomous system numbers?

(A) ICANN (Internet Corporation for Assigned Names and Numbers)
(B) IEEE
(C) W3C
(D) ACM

50. A corporate intranet that connects multiple branch offices over public Internet infrastructure using encryption and tunneling is a:

(A) Virtual Private Network (VPN)
(B) Storage Area Network (SAN)
(C) Public Switched Telephone Network (PSTN)
(D) Metropolitan Ring

51. In the Von Neumann CPU instruction execution cycle, what are the fundamental sequential phases executed for every machine instruction?

(A) Fetch, Decode, Execute, and Store (Write-back)
(B) Compile, Link, Load, and Run
(C) Input, Process, Output, and Terminate
(D) Interrupt, Poll, Trap, and Resume

52. The Program Counter (PC) register in a computer CPU is responsible for:

(A) Holding the memory address of the NEXT instruction to be fetched and executed
(B) Storing the current arithmetic result
(C) Counting total clock cycles elapsed
(D) Holding the opcode being decoded

53. The Instruction Register (IR) holds:

(A) The binary instruction currently being decoded and executed by the Control Unit
(B) The address of the next stack frame
(C) The CPU status flags
(D) The memory base address

54. Which hardware mechanism allows peripheral devices (such as disk controllers or network cards) to transfer blocks of data directly to RAM without CPU intervention?

(A) Direct Memory Access (DMA)
(B) Programmed I/O
(C) Interrupt-driven polling
(D) Virtual memory paging

55. An asynchronous signal transmitted by hardware or software alerting the CPU to suspend current execution and service an urgent event is an:

(A) Interrupt
(B) Trap exception
(C) System bus stall
(D) Cache miss

56. Pipelining in modern CPU microarchitecture boosts instruction throughput by:

(A) Overlapping the execution of multiple consecutive instructions across different hardware stages
(B) Increasing core clock frequency infinitely
(C) Eliminating cache memory entirely
(D) Executing all instructions serially

57. RISC (Reduced Instruction Set Computer) processors differ from CISC (Complex Instruction Set Computer) processors primarily by:

(A) Utilizing fixed-length, single-cycle instructions with load/store architecture and hardwired control
(B) Having hundreds of variable-length complex instructions
(C) Relying heavily on microcoded ROM control units
(D) Having fewer general-purpose registers

58. Which level of CPU cache is typically located directly on the processor core chip, running at full CPU clock speed with lowest latency?

(A) L1 Cache (Level 1)
(B) L2 Cache
(C) L3 Cache
(D) Main DRAM

59. Cache memory operates effectively based on which two fundamental principles of computer program behavior?

(A) Temporal Locality and Spatial Locality
(B) Virtual paging and segmentation
(C) Instruction prefetching and branch prediction
(D) Amdahl's law and Moore's law

60. Temporal Locality implies that:

(A) A memory location referenced recently is highly likely to be accessed again in the near future (e.g., loops)
(B) Memory locations physically adjacent to the current address will be accessed next
(C) Data access time increases linearly with clock frequency
(D) Cache contents expire periodically with real time

61. Spatial Locality implies that:

(A) Memory words stored at physical addresses adjacent to the current address are likely to be referenced soon (e.g., arrays)
(B) Instructions must reside in the same CPU register
(C) Data is distributed across distant cloud servers
(D) Programs execute in random memory spaces

62. A computer architecture with a 32-bit address bus can directly reference a maximum physical memory address space of:

(A) 4 Gigabytes (2^32 bytes = 4,294,967,296 bytes)
(B) 2 Gigabytes
(C) 64 Gigabytes
(D) 1 Terabyte

63. A 64-bit address bus theoretically provides an addressable physical memory capacity of:

(A) 16 Exabytes (2^64 bytes)
(B) 4 Terabytes
(C) 1 Petabyte
(D) 128 Gigabytes

64. In Little-Endian byte ordering architecture (e.g., x86), a multi-byte word (such as 0x12345678) is stored in memory with:

(A) The least significant byte (0x78) at the lowest memory address
(B) The most significant byte (0x12) at the lowest address
(C) Bytes scrambled symmetrically
(D) Higher bits shifted to parity registers

65. In Big-Endian byte ordering architecture (e.g., network byte order in TCP/IP), a multi-byte word is stored with:

(A) The most significant byte at the lowest memory address
(B) The least significant byte at the lowest address
(C) High nibble preceding low nibble
(D) Even bytes in odd banks

66. The standard network byte order universally mandated for multi-byte integers in Internet protocols (IP, TCP, UDP) is:

(A) Big-Endian
(B) Little-Endian
(C) Middle-Endian
(D) Bi-Endian dynamic

67. Which basic logic gate produces an output of HIGH (1) if and only if an ODD number of its binary inputs are HIGH?

(A) XOR Gate (Exclusive OR)
(B) NAND Gate
(C) NOR Gate
(D) AND Gate

68. Which digital logic gates are considered 'Universal Gates' because ANY Boolean logic circuit can be synthesized using them alone?

(A) NAND and NOR Gates
(B) AND and OR Gates
(C) NOT and XOR Gates
(D) XOR and XNOR Gates

69. A combinational digital circuit that performs binary addition of two 1-bit inputs generating a Sum (S) and Carry (C) is a:

(A) Half Adder
(B) Full Adder
(C) Multiplexer
(D) Flip-Flop

70. A Full Adder differs from a Half Adder by:

(A) Accepting three binary inputs: two data bits and an incoming Carry-in bit (C_in)
(B) Handling 4-bit parallel words
(C) Using synchronous clock pulses
(D) Providing negative numbers directly

71. The basic 1-bit bistable digital memory storage element used to construct SRAM and CPU registers is the:

(A) Flip-Flop (or Latch)
(B) Multiplexer
(C) Demultiplexer
(D) Comparator

72. DRAM (Dynamic RAM) cells store each binary bit using a single transistor and a tiny capacitor, requiring:

(A) Periodic electrical refreshing cycles to prevent charge leakage and data loss
(B) Bistable cross-coupled inverters without refresh
(C) Flash laser pulses
(D) Continuous magnetic alignment

73. SRAM (Static RAM) is typically used for CPU caches rather than DRAM because:

(A) SRAM is vastly faster and requires no refresh cycles, though it has lower density and higher cost per bit
(B) SRAM is non-volatile
(C) SRAM consumes zero standby power
(D) SRAM can store multiple bits per capacitor

74. A digital combinational circuit that selects one binary input data line from multiple inputs based on select lines is a:

(A) Multiplexer (MUX)
(B) Demultiplexer (DEMUX)
(C) Decoder
(D) Encoder

75. Moore's Law, an empirical observation made by Gordon Moore in 1965, stated that:

(A) The number of transistors on a microchip doubles approximately every 18 to 24 months
(B) CPU clock speed doubles every 6 months
(C) Network bandwidth exceeds disk capacity by 10x
(D) Software efficiency grows exponentially

76. Amdahl's Law defines the theoretical speedup limit S of an entire system when only a fraction p of the program is parallelized across N processors as:

(A) S = 1 / [(1 - p) + (p / N)]
(B) S = (1 - p) + N * p
(C) S = N / [1 + (N - 1) * p]
(D) S = p * N / (1 - p)

77. If 80% of a computer program (p = 0.80) can be executed in parallel, what is the theoretical maximum speedup limit S as N approaches infinity?

(A) 5.0x (1 / (1 - 0.8) = 1 / 0.2)
(B) 8.0x
(C) 10.0x
(D) Infinite speedup

78. Gustafson's Law modifies Amdahl's perspective for scalable computing workloads by asserting that speedup S with N processors scales as:

(A) S = N - (1 - p) * (N - 1) = (1 - s) + s * N
(B) S = 1 / (1 - p)
(C) S = log2(N)
(D) S = sqrt(N * p)

79. In computer arithmetic, 2's Complement representation of an n-bit signed integer allows addition and subtraction because:

(A) Subtractions A - B are performed by adding A to the 2's complement of B: A + (~B + 1), using identical hardware
(B) It eliminates negative numbers entirely
(C) It provides two distinct representations for zero (+0 and -0)
(D) It requires separate subtractor circuits

80. The 8-bit two's complement representation of the decimal integer - 42 is:

(A) 11010110 (42 = 00101010 => Invert: 11010101 => Add 1: 11010110)
(B) 10101010
(C) 11101010
(D) 10010110

81. The range of signed integers that can be represented in n-bit two's complement format is:

(A) - 2^(n-1) to + (2^(n-1) - 1)
(B) - (2^(n-1) - 1) to + (2^(n-1) - 1)
(C) - 2^n to + 2^n
(D) 0 to 2^n - 1

82. In floating-point arithmetic conforming to IEEE 754 Single Precision (32-bit), the bit allocation is:

(A) 1 Sign bit, 8 Exponent bits (bias 127), and 23 Mantissa (fraction) bits
(B) 1 Sign bit, 11 Exponent bits, 20 Mantissa bits
(C) 2 Sign bits, 6 Exponent bits, 24 Mantissa bits
(D) 4 Sign bits, 8 Exponent bits, 20 Mantissa bits

83. In IEEE 754 Double Precision (64-bit) floating-point format, the bit fields are partitioned into:

(A) 1 Sign bit, 11 Exponent bits (bias 1023), and 52 Mantissa bits
(B) 1 Sign bit, 15 Exponent bits, 48 Mantissa bits
(C) 1 Sign bit, 8 Exponent bits, 55 Mantissa bits
(D) 2 Sign bits, 10 Exponent bits, 52 Mantissa bits

84. In CPU instruction pipelining, a 'Structural Hazard' arises when:

(A) Two or more instructions simultaneously require access to the same physical hardware resource (e.g., single memory port)
(B) An instruction depends on the result of a previous instruction still in the pipeline
(C) A conditional branch alters the instruction flow
(D) The CPU clock frequency drops

85. A 'Data Hazard' (Read-After-Write RAW hazard) in pipelined execution is resolved by which hardware optimization?

(A) Data Forwarding (Bypassing) from execution stage to ALU input
(B) Branch target buffer prediction
(C) Flushing the entire pipeline
(D) Doubling the clock frequency

86. A 'Control Hazard' (Branch Hazard) occurs due to:

(A) Conditional branch instructions where the next instruction address is unknown until branch condition is evaluated
(B) Register write port conflicts
(C) ALU division by zero
(D) Memory bus contention

87. Which hardware architectural technique executes instructions out of program order based on operand availability to avoid stalls?

(A) Out-of-Order Execution with Tomasulo's Algorithm and Reorder Buffer (ROB)
(B) Single-cycle execution
(C) Lock-step synchronous execution
(D) Interrupt vector polling

88. Flynn's Taxonomy classifies computer architectures into four classes based on instruction and data streams: SISD, SIMD, MISD, and:

(A) MIMD (Multiple Instruction, Multiple Data)
(B) SMPD
(C) SPMD
(D) NUMA

89. Modern Graphics Processing Units (GPUs) achieve extreme throughput on graphics and AI matrix computations using which Flynn architecture?

(A) SIMD (Single Instruction, Multiple Data) / SIMT
(B) SISD (Single Instruction, Single Data)
(C) MISD (Multiple Instruction, Single Data)
(D) Serial pipeline

90. In a symmetric multiprocessing (SMP) system, what hardware mechanism ensures that all CPU core caches hold identical data for shared memory locations?

(A) Cache Coherence Protocols (e.g., MESI protocol: Modified, Exclusive, Shared, Invalid)
(B) Virtual memory page tables
(C) Direct Memory Access (DMA)
(D) Write-through DRAM refresh

91. In the MESI cache coherence protocol, what does the 'Exclusive' (E) state designate?

(A) The cache line is present ONLY in this core's cache, is clean (matches main memory), and can be modified without bus notification
(B) The cache line is shared across all cores and read-only
(C) The cache line has been modified and is dirty
(D) The cache line is invalid and must be refetched

92. Nyquist's Sampling Theorem dictates that to perfectly reconstruct an analog signal of maximum frequency B Hz without aliasing, the minimum sampling rate f_s is:

(A) f_s >= 2 * B samples per second
(B) f_s >= B samples per second
(C) f_s >= 4 * B samples per second
(D) f_s >= B / 2 samples per second

93. Nyquist's Maximum Bitrate Formula for a noiseless channel of bandwidth B Hz with M distinct discrete signal levels is:

(A) Bitrate = 2 * B * log2(M) bps
(B) Bitrate = B * log2(M) bps
(C) Bitrate = 2 * B * M bps
(D) Bitrate = B * (M - 1) bps

94. Shannon's Channel Capacity Theorem defines the theoretical maximum error-free data rate C across a noisy channel with Signal-to-Noise Ratio (SNR) as:

(A) C = B * log2(1 + SNR) bps
(B) C = 2 * B * log2(SNR) bps
(C) C = B * (1 + SNR) bps
(D) C = SNR * log2(1 + B) bps

95. A telephone channel has a bandwidth B = 3000 Hz and Signal-to-Noise Ratio SNR = 3162 (SNR_dB = 35 dB). The theoretical Shannon capacity C is:

(A) 34,860 bps (~ 34.86 kbps)
(B) 56,000 bps
(C) 9,600 bps
(D) 128,000 bps

96. A network link transmits 1000 packets per second with average queuing wait time W_q = 0.05 seconds. Little's Law (L = lambda * W) states the average queue length L is:

(A) 50 packets (1000 * 0.05)
(B) 20 packets
(C) 500 packets
(D) 5 packets

97. In store-and-forward packet switching across N consecutive identical router links, the total end-to-end delay for a packet of length L bits with link rate R and propagation delay d_prop per link (zero queueing/processing delay) is:

(A) Total Delay = N * (L / R + d_prop)
(B) Total Delay = (L / R) + N * d_prop
(C) Total Delay = N * L / R
(D) Total Delay = (L * N) / (R * d_prop)

98. If a message of M bits is partitioned into k smaller packets (each of payload P = M/k bits plus header h bits) and sent over N links, packetization achieves:

(A) Pipelining / parallelism across links, drastically reducing end-to-end delay compared to non-packetized store-and-forward
(B) Higher transmission errors
(C) Lower overall network bandwidth utilization
(D) Increased cable resistance

99. A slotted ALOHA network channel with maximum traffic load G = 1.0 achieves a theoretical maximum channel throughput S of:

(A) S_max = 1 / e ~= 0.368 (36.8 %)
(B) S_max = 1 / (2*e) ~= 0.184 (18.4 %)
(C) S_max = 0.50 (50 %)
(D) S_max = 1.0 (100 %)

100. Pure (Unslotted) ALOHA has a vulnerability window of 2*T_frame, resulting in a maximum theoretical throughput of:

(A) S_max = 1 / (2 * e) ~= 0.184 (18.4 %)
(B) S_max = 1 / e ~= 0.368 (36.8 %)
(C) S_max = 0.25 (25 %)
(D) S_max = 0.50 (50 %)

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the Von Neumann Architecture of modern computing systems. Detail its primary structural components, functional registers, and the complete instruction execution cycle.

Q2. Analyze the Von Neumann Bottleneck. Explain its physical origins and detail five major microarchitectural techniques implemented in modern computing systems to mitigate it.

Q3. Define Data Communication and state its five core structural components. Distinguish between Simplex, Half-Duplex, and Full-Duplex transmission modes with authentic computing examples.

Q4. Compare the four fundamental physical Network Topologies: Mesh, Star, Bus, and Ring. Evaluate each with respect to: Link complexity, Cable length, Installation cost, Fault tolerance, and Failure impact.

Q5. Distinguish between Circuit Switching and Packet Switching. Detail the differences in resource reservation, connection establishment, bandwidth utilization, latency characteristics, and resilience to node failures.

Q6. Analyze advanced computing and data communication topic #6: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q7. Analyze advanced computing and data communication topic #7: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q8. Analyze advanced computing and data communication topic #8: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q9. Analyze advanced computing and data communication topic #9: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q10. Analyze advanced computing and data communication topic #10: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q11. Analyze advanced computing and data communication topic #11: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q12. Analyze advanced computing and data communication topic #12: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q13. Analyze advanced computing and data communication topic #13: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q14. Analyze advanced computing and data communication topic #14: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q15. Analyze advanced computing and data communication topic #15: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q16. Analyze advanced computing and data communication topic #16: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q17. Analyze advanced computing and data communication topic #17: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q18. Analyze advanced computing and data communication topic #18: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q19. Analyze advanced computing and data communication topic #19: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q20. Analyze advanced computing and data communication topic #20: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q21. Analyze advanced computing and data communication topic #21: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q22. Analyze advanced computing and data communication topic #22: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q23. Analyze advanced computing and data communication topic #23: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q24. Analyze advanced computing and data communication topic #24: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q25. Analyze advanced computing and data communication topic #25: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q26. Analyze advanced computing and data communication topic #26: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q27. Analyze advanced computing and data communication topic #27: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q28. Analyze advanced computing and data communication topic #28: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q29. Analyze advanced computing and data communication topic #29: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

Q30. Analyze advanced computing and data communication topic #30: Detail the mathematical formulation, protocol standards, architectural tradeoffs, and practical engineering implementation.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[Detailed Step-by-Step Answer]:
1. Physical Origin and Definition:
The Von Neumann Bottleneck refers to the severe throughput limitation caused by the shared physical bus between the CPU and Main Memory.
Over the past four decades, CPU processing speeds have scaled at ~50% per year, whereas DRAM memory latency and bandwidth improved by only ~7% per year.
Consequently, ultra-fast CPUs spend hundreds of idle clock cycles waiting for instructions and data to arrive across the narrow system bus (the 'Memory Wall').

2. Microarchitectural Mitigation Solutions:
- 1. Hierarchical Multi-Level Caches (L1, L2, L3):
Integrates ultra-fast Static RAM (SRAM) directly onto the CPU silicon die operating at processor clock speed.
Exploiting Temporal and Spatial Locality, cache hit rates frequently exceed 95%, satisfying memory requests in 1 to 10 cycles rather than ~200 DRAM cycles.
- 2. Harvard Architecture Separation in Caches (Split L1 Caches):
While external memory remains unified (Von Neumann), on-chip L1 cache is strictly physically partitioned into separate L1 Instruction Cache (L1-I) and L1 Data Cache (L1-D), allowing simultaneous instruction fetching and data operand read/write operations without structural bus conflicts.
- 3. Hardware Prefetching and Branch Prediction:
Sophisticated branch prediction units predict conditional jump paths with >95% accuracy.
Prefetch engines proactively stream sequential cache lines from DRAM into L2/L3 before the CPU explicitly requests them.
- 4. Out-of-Order (OoO) Dynamic Execution:
When an instruction stalls awaiting data from main memory, the CPU's reservation stations and reorder buffer dynamically dispatch and execute subsequent independent instructions, maximizing ALU hardware utilization.
- 5. Direct Memory Access (DMA) and High-Bandwidth Interconnects:
Dedicated DMA controllers offload bulk I/O transfers from the CPU. Modern systems employ point-to-point serial fabrics (PCIe Gen 5, QPI/UPI, Infinity Fabric) and multi-channel DDR5 memory with integrated on-die ECC.

>> Solution to Question 3: Define Data Communication and state its five core structural components. Distinguish between Simplex, Half-Duplex, and Full-Duplex t

3. Transmission Modes Comparison:
- Simplex Mode:
* Directionality: Strictly unidirectional data transfer; channel capacity is permanently dedicated to one direction.
* Mechanism: Device A can only transmit; Device B can only receive.
* Practical Computing Examples: Traditional computer keyboard to CPU motherboard interface, GPS receiver receiving satellite signals, CRT/LCD monitor receiving video signals from GPU.
- Half-Duplex Mode:
* Directionality: Bidirectional communication, but strictly alternating: transmission can occur in either direction, but NEVER simultaneously.
* Mechanism: When Device A transmits, Device B must listen; line turn-around time is required to switch roles.
* Practical Computing Examples: Walkie-talkie radio systems, classical legacy 10BASE2 coaxial Ethernet with CSMA/CD, I2C embedded microcontroller sensor bus.
- Full-Duplex Mode:
* Directionality: Simultaneous bidirectional communication; both parties can transmit and receive data concurrently.
* Mechanism: The physical channel is divided into two separate paths (separate wire pairs or distinct carrier frequencies).
* Practical Computing Examples: Modern switched 1000BASE-T Gigabit Ethernet, standard mobile cellular phone voice calls, bidirectional TCP socket connections.

>>**Solution to Question 4:** Compare the four fundamental physical Network Topologies: Mesh, Star, Bus, and Ring. Evaluate each with respect to: Link complexity

4. Ring Topology:
- Link Complexity: Each node connects to exactly two adjacent neighbors forming a closed loop; requires n links.
- Cable Length & Cost: Moderate cabling; deterministic token-passing protocol prevents packet collisions.
- Fault Tolerance: Inherently low in unidirectional rings (a break anywhere disrupts token circulation).
- Failure Impact: In modern implementations, dual counter-rotating fiber rings (e.g., FDDI, SONET) provide self-healing automatic loopback reconfiguration upon cable cuts.

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models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution: Question 8: Analyze advanced computing and data communication topic #8: Detail the mathematical formulation, protocol standards, architectural t

models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>> Solution to Question 10: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Problem Statement: Analyze advanced computing and data communication topic #13: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution: Question 18: Analyze advanced computing and data communication topic #18: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution to Question 20: Analyze advanced computing and data communication topic #20: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution to Question 24: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution: Question 25: Analyze advanced computing and data communication topic #25: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.

>>Solution Statement: Analyze advanced computing and data communication topic #29: Detail the mathematical formulation, protocol standards, architectural

transmission models, clock synchronization, framing protocols, and bus arbitration.

2. Mathematical & Algorithmic Formulation:
- Link budget calculations and Bit Error Rate (BER) metrics.
- Throughput efficiency: eta = T_frame / (T_frame + 2*T_prop).
- Little's Law queueing model: N = lambda * W.

3. Practical Engineering Application:
Real-world deployment in distributed cloud data centers, fiber optic backbone infrastructure, and embedded microprocessors.
